Foundations of Natural Language Processing (Demonstration Notes)

Introduction to Natural Language Processing

Natural language processing (NLP) is a branch of artificial intelligence that enables computers to analyse, understand and generate human language. NLP combines ideas from linguistics, computer science and statistics. Typical NLP applications include machine translation, sentiment analysis, chatbots, spam filtering and information retrieval.

Human language is difficult for computers because it is ambiguous, context dependent and constantly changing. A single sentence can have several valid interpretations, so NLP systems must use context to choose the most likely meaning.

Early NLP systems relied on hand-written rules, whereas modern systems learn patterns from large collections of text called corpora. A corpus is a large, structured collection of texts used to study language or train models.

Text Preprocessing

Text preprocessing is the process of cleaning and standardising raw text before analysis. Preprocessing usually includes lowercasing, removing punctuation, handling numbers and removing extra whitespace. The quality of preprocessing has a direct effect on every later stage of an NLP pipeline.

Noise removal refers to deleting irrelevant content such as HTML tags, page numbers and repeated headers. Normalisation means converting text into a consistent form, for example expanding contractions or mapping different spellings of a word to one standard spelling.

Tokenization

Tokenization is the process of splitting text into smaller units called tokens. Tokens are usually words, subwords or punctuation marks. Sentence segmentation divides a paragraph into individual sentences, while word tokenization divides a sentence into words.

Tokenization is not trivial because abbreviations such as Dr. and e.g. contain full stops that do not end a sentence. Subword tokenization methods such as Byte Pair Encoding split rare words into frequent smaller pieces, which helps models handle words they have never seen before.

Stop Words, Stemming and Lemmatization

Stop words are very common words such as the, is and and that carry little meaning on their own. Removing stop words reduces the size of the vocabulary and lets an algorithm focus on informative terms.

Stemming is a technique that removes word suffixes using simple rules to obtain a root form. The Porter stemmer is a well-known rule-based stemming algorithm for English.
Lemmatization is the process of reducing a word to its dictionary form, called the lemma, using vocabulary and morphological analysis.

Stemming is fast but can produce non-words such as studi, whereas lemmatization is slower

but returns valid dictionary words such as study.

Part-of-Speech Tagging and Named Entity Recognition

Part-of-speech tagging is the task of assigning a grammatical category, such as noun, verb or adjective, to each token in a sentence. Part-of-speech tags help later stages such as parsing and information extraction.

Named entity recognition (NER) is the task of locating and classifying named entities such as people, organisations, locations and dates in text. Named entity recognition is used in search engines, news analysis and customer support systems to extract structured information from unstructured text.

Hidden Markov Models and Conditional Random Fields are traditional sequence labelling models used for part-of-speech tagging and named entity recognition.

Text Representation: Bag of Words and TF-IDF

Machine learning algorithms require numerical input, so text must be converted into vectors. The bag-of-words model represents a document as a vector of word counts and ignores word order.

Term frequency measures how often a term appears in a document. Inverse document frequency measures how rare a term is across the whole collection of documents. TF-IDF is a weighting scheme that multiplies term frequency by inverse document frequency.

TF-IDF gives high weight to terms that are frequent in one document but rare in the collection, so common words receive low weight. Cosine similarity is a measure of similarity between two vectors based on the angle between them. Cosine similarity is used in information retrieval to rank documents by their relevance to a query.

Word Embeddings

Word embeddings are dense vector representations of words in which words with similar meanings have similar vectors. Unlike bag-of-words vectors, word embeddings capture semantic relationships between words.

Word2Vec is a neural method that learns word embeddings by predicting words from their surrounding context. The distributional hypothesis states that words appearing in similar contexts tend to have similar meanings.

Contextual embeddings produced by transformer models assign different vectors to the same word in different sentences, because the surrounding words change the representation.

Language Models

A language model is a model that assigns probabilities to sequences of words. An n-gram model estimates the probability of a word using only the previous n minus one words.

Smoothing is a technique that assigns a small probability to word sequences that never appeared in the training corpus. Perplexity is a metric that measures how well a language model predicts a sample of text, and lower perplexity indicates a better model.

Neural language models based on the transformer architecture use self-attention to relate every word in a sentence to every other word. Language models are used for text generation,

autocomplete, speech recognition and machine translation.

Evaluation of NLP Systems

Precision is the proportion of predicted positive items that are actually correct. Recall is the proportion of actual positive items that the system successfully finds. The F1 score is the harmonic mean of precision and recall.

Accuracy can be misleading on imbalanced datasets because a model can score highly by always predicting the majority class. A confusion matrix is a table that summarises correct and incorrect predictions for each class.

BLEU is a metric used to evaluate machine translation by comparing system output with reference translations.